

$^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances **1967Ko19,1977Ou01,1981Bi05** $^{34}\text{S}(\text{p,p}'\gamma)$ measurement:

1967Ko19: E=2.78-3.08-MeV proton beams were produced from the 4-MeV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences, in energy step of 1 keV. Target was enriched thin ^{34}S . γ rays were detected with a 70 by 60 mm NaI(Tl) crystal for detecting the 2.13 MeV γ -ray of ^{34}S decay, resolution of 11.5% for the 0.622 MeV ^{137}Cs line. Measured $\sigma(E_p, \theta)$ from yields of the 2.13 MeV γ -ray, $\text{py}(\theta)$ of the 2.13 MeV γ -ray. Deduced resonance energies, J, π , relative resonance intensities.

 $^{34}\text{S}(\text{p,p})$ measurements:

1977Ou01: E=1.45-2.82 MeV proton beams were produced from the TUNL 3-MV Van de Graaff accelerator in energy steps of 100 eV for on-resonance and 400 eV for off-resonance, overall FWHM=350-450 eV. Target was CdS, with 90% enriched ^{34}S evaporated onto 5-10 $\mu\text{g}/\text{cm}^2$ carbon backings. Scattered protons were detected with surface barrier detectors at 90°, 105°, 135° and 160°. Measured $\sigma(E_p, \theta)$. Deduced resonance energies, J, π , Γ_p from R-Matrix analysis.

1981Bi05: E=1.4-2.8 MeV proton beams were produced from the Groningen 5 MV Van de Graaff accelerator. Targets were 4 and 1 $\mu\text{g}/\text{cm}^2$ Sb_2S_3 (90% in ^{34}S) evaporated onto 10 $\mu\text{g}/\text{cm}^2$ carbon backings. Scattered protons were detected with three silicon surface barrier detectors at 88°, 124° and 174°. Measured $\sigma(E_p, \theta)$. Deduced resonance energies, J, π , Γ_p . Also measured (p, γ).

1971BrXT: Measured $\sigma(E_p)$. Deduced resonance energies, J, π , widths.

1974Bi16: E=1.2-2.1 MeV protons were produced from the Helsinki University 2.5 MV van de Graaff accelerator. Targets were prepared from natural sulphur (4.22% ^{34}S) with ^{34}S ions embedded into carbon foils. Scattered protons were detected with four silicon surface barrier detectors. Measured $\sigma(E_p, \theta)$. Deduced resonance energies, J, π , widths from analysis of the resonances.

Others: **1978Ce02**, **1980Fa07**, **1983Ch50**, **1988Za04**, **2019Ka53**.

 ^{35}Cl Levels

A_2 and A_4 values under comments are from $\text{py}(\theta)$ of the 2.13 MeV γ -ray from ^{34}S (**1967Ko19**).

I(res) under comments are resonance intensities relative to 1 for $E_p(\text{lab})=2799$ resonance (**1967Ko19**).

E(level) [†]	J ^π [‡]	Γ_p [#]	$E_p(\text{lab})(\text{keV})$ [#]	Comments
7195@	1/2 ⁻ @		848	$\Gamma=27$ eV 8 (1971BrXT).
7274@	1/2 ⁻ @		929	$\Gamma=14$ eV 4 (1971BrXT).
7550	7/2 ⁻		1214	$\Gamma_p=4.6$ eV 10 and $\Gamma=5$ eV +4-2 from a Q^2 minimization analysis of the (p,p) excitation function (1971BrXT). $\Gamma=10.5$ eV 15, $\Gamma_p=5.8$ eV 7, $\Gamma_\gamma=4.7$ eV 17 (1971BrXT); $\Gamma=11.0$ eV 15, $\Gamma_p=7.2$ eV 15, $\Gamma_\gamma=3.8$ eV 15 (1974Bi16), which are deduced using a questionable (p, γ) resonance strength of 21 eV 3 from 1966En04 .
7686&	3/2 ⁻ &		1354	$\Gamma_p=445$ eV 11 for $J^\pi=3/2^-$ (1981Bi05).
7706&	3/2 ⁺ , 5/2 ⁺ &		1375	$\Gamma_p=4$ eV 1 for $J^\pi=5/2^+$, 17 eV 3 for $J^\pi=3/2^+$ (1981Bi05).
7796 3	(1/2) ⁻	0.031 keV 10	1467.3 5	
7837 3	3/2 ⁻	3.7 keV 4	1510.0 5	$\Gamma=3.8$ keV 2, $\Gamma_p=3.8$ keV 2 (1974Bi16).
7880 3	3/2 ⁺ , 5/2 ⁺	0.008 keV 4	1554.0 5	
8004 3	3/2 ⁺ , 5/2 ⁺	0.011 keV 5	1682.1 5	$\Gamma_p=21$ keV 3 for $J^\pi=5/2^+$, 24 eV 3 for $J^\pi=3/2^+$ (1981Bi05).
8034 3	1/2 ⁻ , 3/2 ⁻	0.026 keV 7	1712.9 5	
8038 3	1/2 ⁺	0.30 keV 2	1716.9 5	
8148 3	1/2 ⁻	2.66 keV 27	1830.2 5	
8150 3	3/2 ⁻	0.56 keV 6	1831.5 5	
8209 3	(5/2) ⁺	0.033 keV 10	1893.2 5	$\Gamma=100$ eV 20, $\Gamma_p=120$ eV 20 (1974Bi16).
8211 3	1/2 ⁺	0.094 keV 15	1894.4 5	$\Gamma_p=39$ eV 5 for $J^\pi=5/2^+$, 44 eV 6 for $J^\pi=3/2^+$, 120 eV 11 for $J^\pi=1/2^+$ (1981Bi05).
				$\Gamma=270$ eV 50, $\Gamma_p=270$ eV 50 (1974Bi16).
8216	5/2 ⁺	0.014 keV 3	1900	$\Gamma=14$ eV 4, $\Gamma_\gamma=0.78$ eV (1974Bi16).
8243 3	3/2 ⁻	0.140 keV 15	1928.2 5	
8271 3	3/2 ⁺ , 5/2 ⁺	0.005 keV 3	1956.1 5	
8279 3	3/2 ⁺ , 5/2 ⁺	0.006 keV 3	1964.6 5	
8290 3	(1/2) ⁻	0.04 keV 1	1975.7 5	$\Gamma_p=49$ eV 11 for $J^\pi=3/2^-$, 48 eV 6 for $J^\pi=1/2^-$ (1981Bi05).

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$^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma):\text{resonances}$ **1967Ko19,1977Ou01,1981Bi05 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J^π [‡]	Γ_p [#]	$E_p(\text{lab})(\text{keV})$ [#]	Comments
8300 3	3/2 ⁻	0.073 keV 15	1986.7 5	$\Gamma=30$ eV 7, $\Gamma_p=28$ eV 6, $\Gamma_\gamma=2.4$ eV (1974Bi16). J^π : 5/2 ⁺ in 1974Bi16.
8383 3	3/2 ⁺ , 5/2 ⁺	0.023 keV 7	2072.2 5	
8404 3	5/2 ⁻ , 7/2 ⁻	0.002 keV 1	2093.9 5	$\Gamma_p=4$ eV 1 for $J^\pi=7/2^-$, 2 eV 1 for $J^\pi=5/2^-$ (1981Bi05).
8406 3	(5/2 ⁻ , 7/2 ⁻)	0.001 keV 1	2095.2 5	
8409 3	1/2 ⁻	0.125 keV 15	2098.7 5	$\Gamma_p=59$ eV 7 for $J^\pi=5/2^+$, 67 eV 8 for $J^\pi=3/2^+$ (1981Bi05). $I(\text{res})=1$. $E_p(\text{lab})(\text{keV})$: from 1967Ko19.
8418 3	1/2 ⁺	0.026 keV 7	2107.5 5	
8435 3	(3/2) ⁺	0.090 keV 15	2125.9 5	$I(\text{res})=0.18$.
8486 3	3/2 ⁺ , 5/2 ⁺	0.012 keV 5	2178.2 5	
8488 3	3/2 ⁻	0.150 keV 15	2179.7 5	$I(\text{res})=0.34$.
8515 3	1/2 ⁻	0.150 keV 15	2208.1 5	
8573 3	(5/2) ⁺	0.08 keV 1	2267.9 5	$I(\text{res})=0.31$.
8582 3	1/2 ⁺	0.75 keV 8	2276.7 5	
8592 3	3/2 ⁺ , 5/2 ⁺	0.003 keV 2	2287.4 5	$I(\text{res})=0.16$.
8616 3	5/2 ⁺	0.175 keV 20	2311.6 5	
8620 3	3/2 ⁺ , 5/2 ⁺	0.002 keV 1	2316.2 5	$I(\text{res})=0.34$.
8633 3	(3/2, 5/2 ⁺)	0.001 keV 1	2328.8 5	
8643 3	3/2 ⁺ , 5/2 ⁺	0.003 keV 2	2339.9 5	$I(\text{res})=0.27$.
8687 3	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	2385.0 5	
8689 3	1/2 ⁺	0.20 keV 2	2387.2 5	$I(\text{res})=0.29$.
8691 3	1/2 ⁻	6.4 keV 7	2388.6 5	
8698 3	3/2 ⁻	0.8 keV 1	2396.0 5	$I(\text{res})=3.70$.
8750 3	3/2 ⁻	0.30 keV 3	2449.5 5	
8775 3	1/2 ⁻	0.57 keV 6	2475.2 5	$I(\text{res})=0.16$.
8781 3	3/2 ⁻	0.214 keV 25	2481.7 5	
8788 3	(3/2 ⁺ , 5/2 ⁺)	0.001 keV 1	2489.1 5	$I(\text{res})=0.34$.
8799 3	(3/2 ⁺ , 5/2 ⁺)	0.001 keV 1	2500.3 5	
8825 3	1/2 ⁺	1.70 keV 17	2526.8 5	$I(\text{res})=0.27$.
8829 3	1/2 ⁻	12.2 keV 12	2530.8 5	
8830 3	1/2 ⁺	0.080 keV 15	2532.0 5	$I(\text{res})=0.29$.
8839 3	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	2541.7 5	
8858 3	3/2 ⁺ , 5/2 ⁺	0.010 keV 5	2560.8 5	$I(\text{res})=0.27$.
8870 3	3/2 ⁺ , 5/2 ⁺	0.027 keV 10	2573.2 5	
8909 3	3/2 ⁺ , 5/2 ⁺	0.002 keV 1	2613.2 5	$I(\text{res})=0.29$.
8955 3	(3/2) ⁺	0.075 keV 15	2661.3 5	
8959 3	(1/2 ⁻ , 3/2 ⁻)	0.04 keV 1	2665.4 5	$I(\text{res})=0.27$.
8983 3	5/2 ⁻ , 7/2 ⁻	0.003 keV 2	2690.0 5	
8985 3	3/2 ⁺ , 5/2 ⁺	0.025 keV 10	2692.1 5	$I(\text{res})=0.27$.
8998 3	5/2 ⁻ , 7/2 ⁻	0.002 keV 1	2705.1 5	
9021 3	3/2 ⁻	3.50 keV 35	2728.4 5	$I(\text{res})=0.27$.
9031 3	(5/2) ⁺	0.04 keV 1	2739.4 5	
9040 3	1/2 ⁻	0.292 keV 30	2748.2 5	$I(\text{res})=0.27$.
9048 3	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	2756.7 5	
9050 3	(5/2) ⁺	0.095 keV 15	2758.8 5	$I(\text{res})=0.27$.
9084 3	(5/2) ⁺	0.057 keV 10	2793.2 5	
9089 4			2799 4	$I(\text{res})=0.27$.
9101 3	3/2 ⁻	0.20 keV 2	2810.9 5	
9108 4			2818 4	$I(\text{res})=0.27$.
9114 4			2825 4	
9120 4			2831 4	$I(\text{res})=0.27$.
9127 4			2838 4	
9136 4			2847 4	$I(\text{res})=0.27$.
9147 4			2859 4	
9159 4			2871 4	$I(\text{res})=0.27$.
9168 4	(3/2 ⁻)		2880 4	

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$^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma):\text{resonances}$ **1967Ko19,1977Ou01,1981Bi05** (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab)(keV) [#]	Comments
9185 4		2898 4	A ₂ =+0.19 133, A ₄ =+0.017 1480 (1967Ko19). I(res)=0.11.
9196 4		2909 4	
9207 4	9/2 ⁺	2920 4	I(res)=1.70. A ₂ =+0.09 18, A ₄ =+0.37 12 (1967Ko19).
9222 4		2936 4	I(res)=0.25.
9227 4	5/2 ⁻	2941 4	I(res)=0.59. A ₂ =+0.59 28, A ₄ =+0.29 32 (1967Ko19).
9241 4	7/2 ⁺	2955 4	I(res)=2.32. A ₂ =+0.24 40, A ₄ =+0.27 53 (1967Ko19).
9255 4		2970 4	I(res)=0.45.
9271 4		2986 4	I(res)=0.34.
9276 4		2991 4	I(res)=0.32.
9284 4	5/2 ⁻	3000 4	I(res)=3.86. A ₂ =+0.72 28, A ₄ =-0.17 29 (1967Ko19).
9294 4		3010 4	I(res)=0.68.
9299 4		3015 4	I(res)=0.45.
9325 4	5/2 ⁺	3042 4	I(res)=1.81. A ₂ =+0.81 28, A ₄ =-0.48 35 (1967Ko19).
9336 4	7/2 ⁺	3053 4	I(res)=1.00. A ₂ =+0.17 29, A ₄ =+0.34 40 (1967Ko19).
9345 4		3062 4	I(res)=0.50.
9349 4		3067 4	I(res)=0.50.
9355 4		3073 4	I(res)=2.23.

[†] From E(level)=E_p(cm)+S(p)(^{35}Cl), where S(p)=6370.81 4 (2021Wa16) and E_p(cm) deduced from E_p(lab) by
E_p(cm)=E_p(lab)*[m(^{34}S)/(m(^{34}S)+m(p))].

[‡] From comparison of experimental differential cross sections with theoretical predictions by R-Matrix calculations (1977Ou01) for levels up to 9101 level and from $\text{py}(\theta)$ (1967Ko19) for levels after that.

[#] From 1977Ou01 up to 9101 level and from 1967Ko19 after that, unless otherwise noted. 1977Ou01 state that although the absolute proton energies are accurate only to about 3 keV, the relative values of E_p(lab) should be reliable to within a few hundred eV, based on which the evaluators have assigned a 0.5 keV uncertainty for E_p(lab) and 3 keV uncertainty for E(level) from 1977Ou01.

@ From 1971BrXT.

& From 1981Bi05.